MILITARY SPECIFICATION

MILLING MACHINES, BED TYPE, HORIZONTAL SPINDLE, AUTOMATIC CYCLE, 24, 26 AND 30 INCH MINIMUM TABLE WIDTH, PLAIN AND DUPLEX MACHINES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic cycling, bed type, horizontal machines, with minimum table widths of 24, 26, and 30 inches and table lengths from 72 to 192 inches, plain and duplex spindles, with equipment and accessories as described herein.

1.2 Classification. Milling machines covered by this specification shall be of the types, classes, and sizes as specified (see 6.2).

Type I - Single spindle machines (plain)
Type II - Two spindle machines (duplex)

class 1- 24 inches minimum table width

Size 1 - 72 inches minimum table length
Size 2 - 96 inches minimum table length
Size 3 - 120 inches minimum table length
Size 4 - 144 inches minimum table length
Size 5 - 168 inches minimum table length

Class 2 - 26 inches minimum table width

Size 1 - 96 inches minimum table length
Size 2 - 120 inches minimum table length
Size 3 - 144 inches minimum table length
Size 4 - 168 inches minimum table length
Size 5 - 192 inches minimum table length

class 3 - 30 inches minimum table width

Size 1 - 96 inches minimum table length
Size 2 - 120 inches minimum table length
Size 3 - 144 inches minimum table length
Size 4 - 168 inches minimum table length
Size 5 - 192 inches minimum table length

FSC 3417

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2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 2139 Wisconsin Avenue, N. W., Washington, D. C. 20007.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION STANDARDS

MG 1 Motors and Generators

(Copies may be obtained from the National Electrical Manufacturers' Association, 155 East 44th Street, New York, New York 10017.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360.01

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, One Thomas Circle, N. W., Washington, D. C. 20005.)

AMERICAN NATIONAL STANDARDS

ANSI B5.1 T-S lots - Their Bolts, Nuts, Tongues and Cutters

ANSI B5.18 Spindle Noses and Arbors for Milling Machines

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

3.1 First Article approval. If First Article approval is required (see 6.2 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.2 Desire. The machine shall be one of the manufacturer's current models, new and unused. The machine shall be capable of milling operations in the longitudinal and vertical planes in accordance with requirements specified herein. The machine design and construction shall include all components,

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parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Type I. The plain machine shall have one vertical floor mounted column fixed to one side of the bed. The column shall have vertical ways for mounting the spindle carrier. The spindle carrier shall mount on the column ways with the spindle perpendicular to the longitudinal movement of the table and parallel to the table surface. Machine motions shall be: (1) longitudinal table movement, (2) spindle quill cross movement parallel to the table surface, and (3) vertical spindle carrier movement on the vertical column ways.

3.2.2 Type II. The duplex machine shall have two vertical floor mounted columns fixed to the sides of the bed. Each column shall have vertical ways for mounting the spindle carriers. The spindle carriers shall mount on the column ways with the spindle perpendicular to the longitudinal movement of the table and parallel to the table surface, Machine motions shall be: (1) longitudinal table movement, (2) spindle quill cross movement parallel to the table surface, and (3) vertical spindle carrier movement on the vertical column ways.

3.2.3 Controls. All operating controls and all manual adjustments shall be grouped in a location convenient to the operator, All handwheels and control levers shall automatically and positively disengage during power rapid traverse or power feeds.

3.2.4 Safety devices. The machine shall provide covers, or other safety devices, for all parts that present safety hazards. The devices shall include guards for belts, pulleys, chains, gears, and sprockets. All guards shall provide easy access to the guarded parts and shall not interfere with operation of the machine. A safety device shall be furnished to prevent simultaneous quill advance and table feed. Spindle(s) shall be equipped with a protective device which will stop all slide motions during spindle(s) stall or loss of drive due to malfunction. When special safety devices are required, they shall be as specified (see 6.2).

3.2.5 Lubrication. All bearings, gears, journals, spindles, ways and mating sliding parts shall be provided with means of lubrication. The reservoirs of lubricating systems shall be equipped with oil level sight gages. Recirculating systems shall include filters which may be cleaned or replaced. Ways shall be lubricated either by automatic, metered or one-shot force feed lubricating system. Transmissions and enclosed gear trains shall be flood lubricated or equal. Motors shall be equipped with sealed-for-life lubricated type bearings. All other parts of the machine shall have automatic force feed lubrication.

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3. 2.6 Interchangeability. To provide for replacement of worn parts, all parts produced by the same manufacturer and bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3 Construction. The machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. Processes such as welding, peening, plugging, or filling with solder or metallic paste shall not be used for reclaiming any defective part for use in the machine.

3.3.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer’s standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer’s standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Dials. All dials shall have a positive retention system to assure retention of the dial at the desired setting. All graduations, numerals and letters shall be permanently and legibly displayed and shall have a nonglare finish. Each dial graduation shall represent not more than 0.001 inch of movement.

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3.3.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language. Letters shall be engraved, etched, embossed or stamped in bold-face on a contrasting background.

3.3.9 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.01.

3.4 Components. The milling machine furnished under this specification shall have components as described herein.

3.4.1 Table bed. The bed shall be a casting of iron, iron alloy, or fabricated from alloy steel. The bed shall contain the feed drive mechanism for the table and have necessary access doors or covered openings to facilitate required adjustments or replacement of parts. The accuracy-of the machining and fitting of the bed ways shall be such that the machine will meet all performance requirements stated herein and conform to the alignment tolerances in Table II. An adequate number of leveling screws shall be provided to enable the machine to be properly leveled. The bed shall have ways conforming to 3.4.3.

3.4.2 Table. The table shall be a casting of iron, iron alloy, or alloy steel and accurately machined. The table shall have ways conforming to 3.4.3. The mounting surface of the table shall be provided with not less than the number of T-slats specified in Table I and be machined in accordance with ANSI B5.1. The front face of the table shall have a finished surface with T-slot(s) running the full length of the table for the purpose of mounting adjustable table stops. The T-slot(s) shall be parallel to the longitudinal travel of the table. The table shall have coolant drain channels with chip screens and provisions for returning the coolant to the reservoir. The table shall be provided with power feed and rapid traverse feed for longitudinal table movements. The table shall be fitted to the bed ways and adjusted to required alignment tolerances and accuracies specified in Table II. The table ways shall be automatically lubricated by a pressurized lubricating system capable of providing a film of oil over the entire sliding surface of the ways. The minimum size and travel of the table shall not be less than specified in Table I.

3.4.2.1 Worktable drive system. The table shall be driven by a rack and gear or feed screw located beneath the table and shall be provided with an automatic anti-backlash mechanism. The number of feed rates and range of feed rates shall not be less than specified in Table I. Power rapid traverse independent of feed change shall be provided for either direction of

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longitudinal table travel. The rapid traverse shall not be less than specified in Table I. The table drive mechanism shall be capable of smoothly accelerating, driving, decelerating and reversing the table without shock or sluggish action at the table trip points, when operating at any table speed or rapid traverse speed.

3.4.2.2 Table controls. Table controls necessary for manual and automatic operation shall be mounted on a panel or pendant and clearly identified. Longitudinal and rapid traverse function, forward and reverse directions, shall be controlled by electro-manual controls.

3.4.2.3 Automatic cycle table control. The machine shall be equipped with automatic cycling table controls to actuate rapid traverse and pre-selected feed rate at any point in either direction of the table travel. Automatic reversal of table feed shall be provided.

3.4.3 Ways and way slides. The ways of the machine shall be machined, handscraped, or hardened and ground. The accuracy of the machining and fitting of ways and surfaces shall be such that the machine will meet all performance requirements stated herein and conform to the alignment tolerances in Table II. The machine shall be provided with a simple means of adjustment to insure correct alignment for mating components and compensate for wear of the mating surfaces. Effective way wipers or guards shall be provided on table ways to prevent entrance of chips, dirt or other extraneous material.

3.4.4 Column(s). The machine shall be furnished with the number of columns as required (see 3.2.1 and 3.2.2). The column shall be either a casting of iron, iron alloy, semi-steel or steel weldment. The column(s) shall be fitted to the foundation and the base of the table so that when the spindle head is fitted to the column, the spindle axes will be parallel with the top surface of the worktable and perpendicular to the T-slots of the worktable. The column and spindle shall be fitted and aligned to comply with the tolerances stated in Table II. When vertical power feed is required (see 3.4.5), the column(s) shall be provided with T-slot(s) running vertically along the column for mounting adjustable stops. The column(s) shall have vertical ways conforming to 3.4.3. The column(s) shall be provided with access doors or covered openings as required for servicing components located therein. All leveling screws required by the foundation shall be furnished with the machine.

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3. 4.5 Spindle carrier(s). The spindle carrier(s), for the purpose of this specification, is defined as a self-contained assembly of the machine that houses the quill, spindle, drive motor, transmission, associated mechanism, and the arbor support. The spindle carrier(s) slides and mating ways shall conform to 3.4.3. The spindle carrier(s) shall be constructed to assure acceptable performance throughout the full range and capacity of the machine as specified in Table I and shall comply with tolerances stated in Table II. One spindle carrier for each column shall be furnished when Type II machine is required (see 1.2). Each spindle carrier shall be controlled independently of the other. The feed screws for the vertical travel of the spindle carrier(s) shall be mounted on precision-type ball or roller bearings. The spindle carrier(s) shall be furnished with a hand control having a direct reading open-faced dial for indicating vertical movement. When specified (see 6.2), the spindle carrier shall be furnished with power elevation and power rapid traverse.

3.4.5.1 Spindle(s). Each spindle shall be mounted on precision ball or roller bearings in its spindle carrier. The spindle shall be furnished with precision ground Number 50 National Standard taper in accordance with ANSI B5. 18. A quill shall be provided and, when specified (see 6.2), shall be furnished with automatic retraction. The optional quill adjustment range required shall be specified, (see 6.2), and shall be in accordance with Table I. A graduated measuring system, assembled without backlash shall be provided for positioning of the quill. Spindle controls shall be furnished to start, stop, jog, and reverse the spindle. Optional spindle speed shall be specified, (see 6.2), in accordance with Table I. The spindle shall have an automatic braking mechanism to stop the spindle when power to the spindle motor is turned off. The spindle drive motor shall be furnished with a safety device to insure that power feed is automatically stopped whenever the spindle is stopped. The spindle shall conform to the alignment tolerances and accuracies stated in Table II. The optional spindle horsepower shall be stated in the ordering data (see 6.2).

3.4.6 Overarms and support. Type I, (see 1.2), machine shall be furnished with overarms and supports. The overarms shall have a manual overarm control for positioning. The overarms shall be mated with a sliding surface of dovetail or square way type, or the overarms which shall be of the round bar type mated into the spindle carrier(s). When specified (see 6.2), each spindle carrier for Type II machine shall be furnished with an overarm and supports.

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3. 4.7 Clamping devices. Levers or bolts for clamping the quill(s) and arbors shall be furnished. The spindle carrier(s) shall have clamping devices to clamp the carrier(s) to the column(s) when the carrier(s) is not equipped with power vertical feed.

3.4.8 Safety stops. Means shall be provided to prevent overtravel during power feed or rapid traverse of the table or hand feed of the spindle carrier. Safety devices shall be provided to prevent damage to the feed or rapid traverse mechanisms if the moving parts are inadvertently fed or rapid traversed into the safety stops.

3,5 Sizes and capacities. Sizes and capacities of the milling machines covered by the specification shall be not less than requirements set forth in Table I.

3.5.1 Vertical spindle travel and distance of centerline of spindle to table top. The vertical spindle travel and maximum distance of centerline of spindle to table top, Option A, B, C or D, shall be specified (see 6.2). The option selected for spindle centerline distance to table top shall correspond to the same option for vertical travel. (Example: When Option A is selected for vertical travel, the option for spindle centerline to table top distance will automatically be A.)

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TABLE I

CHARACTERISTICS	CLASS 1					CLASS 2					CLASS 3				
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 1	Size 2	Size 3	Size 4	Size 5	Size 1	Size 2	Size 3	Size 4	Size 5
Table size, minimum, work surface Width by length, inches	24x72	24x96	24x120	24x144	24x168	26x96	26x120	26x144	26x168	26x192	30x96	30x120	30x144	30x168	30x192
Travel range, inches															
Longitudinal	48	72	96	120	144	72	96	120	144	168	72	96	120	144	168
Vertical (see 3.5.1)															
Option A	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
Option B	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
Option C	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
Option D	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26
Quill adjustment, inches (see 3.4.5.1)															
Option A	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Option B	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Feed rate, minimum number of	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
Feed rate, maximum low, minimum high, IPM															
Longitudinal	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150	1-150
Vertical (see 3.4.5)	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20	1-20
Automatic quill retraction, inches, not less than (see 3.4.5.1)															
Option A	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/32
Option B	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32	5/32
Power rapid traverse, not less than, IPM															
Longitudinal	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300
Vertical (see 3.4.5)	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20
Centerline of spindle to table top, inches, maximum, not less than															
Option A (see 3.5.1)	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
Option B	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18
Option C	26	26	26	26	26	26	26	26	26	26	26	26	26	26	26
Option D	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
Minimum, not more than	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Spindle motor, horsepower, not less than															
Option A (see 3.4.5.1)	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
Option B	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20
Option C	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
Option D	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40
Option E	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
Spindle speed, minimum number of	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
Spindle speed, maximum low, minimum high, RPM															
Option A (see 3.4.5.1)	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800	22-800
Option B	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200	30-1200
Option C	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000	50-2000
Option D	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000	75-3000
T-slots, number and size	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16	3-13/16

Requirements are not less than shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.6 _____ Each machine shall be capable of satisfactory performante when tested in accordance with 4.5.1 through 4.5.5.

3.7 Alignment tolerances. Each machine shall meet alignment tolerances of Table II.

TABLE II

ALIGNMENT TOLERANCES	
TEST	TOLERANCES
Spindle(s) end periphery runout	0.0004" TIR
Spindle(s) end face runout	0.0002" TIR
Spindle(s) internal taper runout	
1 inch from spindle(s) nose	0.0004" TIR
12 inches from spindle(s) nose	0.0010" TIR
Bore of arbor support alignment with spindle axis	
12 inches from spindle(s) nose	0.001" TIR
Longitudinal rise and fall of table	0.0005" in any 12-inch span. Total not to exceed +0.0005 plus 0.00015 per 12 inches of travel.
Table top parallel with spindle(s)	0.0010" per foot
Vertical movement of spindle square with table surface	+0.0005" in any 12-inch span. Total not to exceed +0.0005 plus 0.00015 per 12 inches of travel.
Center T-slots parallel to table movement	0.0005" in any 12-inch span. Total not to exceed +0.0005 plus 0.00015 per 12 inches of travel.
Spindle square with center T-slot	0.0005" per foot
Tracer arm parallel to T-slots (where applicable)	0.001" per foot
Tracer bracket parallel to table top (where applicable)	0.001" per foot

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3.8 Electrical equipment. Unless otherwise specified (see 6.2), all electrical components and their locations shall conform to JIC Electrical Standards for General purpose Machine Tools. Undervoltage protection shall be provided even though it may not be required in the JIC Electrical Standards. An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards.

3.8.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower rating shall conform to the requirements shown in Table I. Unless otherwise specified (see 6.2), the machine shall be designed to operate from a 230/460 volt, 3 phase, 60 Hz power supply. The motor shall be initially wired for operation on the voltage specified (see 6.2). Unless otherwise specified (see 6.2), each motor shall meet the requirements of a dripproof enclosure.

3.9 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit the full flow of coolant, and shall have means for draining and cleaning. The system shall include a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant flow.

3.10 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump and on the low side of each pressure reducing valve to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.11 Electrical load indicator. Unless otherwise specified (see 6.2), each spindle drive motor shall have a load meter indicating either amperes or percent full rated load current. Meters shall be accurate within 2% at rated full load current, The indicator shall be clearly readable from the operators normal position.

3.12 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished with each machine:

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2 arbor supports, Style B (Type I machines only)

1 arbor draw-in rod (Two sets when Type II machine is required)

1 set of pick-off spindle speed change gears (Two sets when Type 11 machine is required)

1 set of pick-off table feed change gears (Furnished only when machine function is required)

1 set of trip dogs

1 set of hand tools normally furnished by the manufacturer

3.13 Optional ecluipment. When required (see 6.2), the following optional equipment and accessories shall be furnished.

3.13.1 Automatic program control. When specified (see 6.2), the machine shall be provided with an automatic program control capable of directing longitudinal table movement in either direction at rapid traverse rate or feed rate, quill movement either in or out, and rise-and-fall action on the spindle carrier. The program control shall have means for programming each cycle with all directions of movements and rate of travel.

3.13.2 Tracer system. When required (see 6.2), a tracer system shall be furnished. The tracing mechanism shall be hydraulically operated and capable of controlling machine movement for precise duplication of templates. One-to-one relationships between the template and the workpiece shall be maintained on finish cuts at all feed rates. The tracer mounting bracket shall be attached to the table. The bracket shall be adjustable longitudinally to accurately position the tracer follower to the template in relation to the cutter. The tracer stylus shall respond to pressure light enough to permit the use of wood, plastic or other easily worked template materials. The tracing head shall be of the automatic depth control type and shall provide for disengagement of the tracer mechanism when hand control of the table and spindle carrier movement is required for conventional milling or setup purposes. When tracer system is furnished, it shall comply with tolerances specified in Table II.

3.13.3 Additional optional accessories and equipment. Other optional accessories and equipment shall be furnished when specified (see 6.2).

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3.14 Electromagnetic interference control. Unless otherwise specified (see 6.2), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

3.15 Fungus control. When required (see 6.2), fungus proofing shall be as specified.

3.16 Lubrication chart or plate. Unless otherwise specified (see 6.2), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification Number (if available)

3.17 Nameplate. Unless otherwise specified (see 6.2), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

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Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

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3.18 Identification plate. When specified (see 6.2), a separate identification plate of corrosion-resistant metal permanently and legibly marked in letters 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely attached to each machine. The plate shall have blank space for the government to stamp 25 characters of the same size as the caption.

3.19 Technical data. Data shall be furnished as specified (see 6.2.2).

3.20 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3. 1, First Article inspection shall be performed. Unless otherwise specified (see 6.2), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance.

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Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.5 and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.20,

4.5 Tests.

4.5.1 Operational test The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and accessories shall be verified during the trial period.

4.5.2 Alignment accuracy test. The machine shall be tested to determine that it complies with the alignment accuracy requirements of 3.7,

4.5.3 Performance test. The following performance test shall be performed to determine the structural stability. The machine shall be used to mill a workpiece of SAE 1045 steel. The workpiece shall be not less than 12 inches long by 6 inches wide and 6 inches thick. Scrap blocks may be utilized if available. The test block shall be ground flat on one vertical side and one horizontal side. The horizontal side of the test block previously ground shall be mounted on the table surface with the ground vertical side opposite the spindle nose and in precise alignment with the nearest exposed T-slot. When a duplex machine is required, a workpiece for each spindle shall be furnished. The workpiece may be drilled to facilitate clamping, The machine shall make roughing cuts to remove metal from the horizontal face and the vertical face of the workpiece. Finishing cuts shall be made on the workpiece surfaces which were previously rough cut. The test tolerances of the workplaces after the finishing cuts shall be as shown in 4.5.3.1 through 4.5.3.3.

4.5.3.1 Squareness. The two milled sides shall be checked for squareness. The sides shall be square with each other within 0.001 inch.

4.5.3.2 Flatness. The two milled sides shall be tested for flatness. The sides shall be flat within 0.001 inch.

4.5.3.3 Parallelism. The horizontal and vertical finished surfaces of the workpiece shall be checked for parallelism. The surfaces shall be parallel to the ground surfaces within 0.001 inch.

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4. 5.4 Metal removal and tool chatter test. A metal removal and tool chatter test shall be performed consisting of a roughing cut on the horizontal surface of the workpiece, using a high speed steel slab mill, with loads and speed as shown in Table 111. Axial adjustment of the spindle and arbor shall be suitable for continuous operation of the machine. The electrical load during test may be determined at the discretion of the supplier by the load meter specified in 3. 11 or by a voltmeter and ammeter with accuracy within two percent. The resultant metal removal rate shall be not less than 0.65 cubic inch per minute per horsepower used in the cut. The surface of the workpiece, after the cut, shall show no evidence of tool chatter.

TABLE III

LOADS, SPINDLE SPEEDS, FEED RATES FOR CUTTING AND TOOL CHATTER TEST

Spindle Drive HP	Percent Load	Cutter Size, Inches		Spindle Speed Rev. Per Min.*	Feed Rate In. Per Min.*
		Width	Diameter		
10	100	6	4	75	7.5
15	90	6	4	75	7.5
20	80	6	4	75	7.5
30	70	6	4	75	7.5

* Variation of plus or minus 10 percent in speed and feed rate permissible. Use of 2 cutters permissible in tests of 20 and 30 HP spindle drives.

4,5.5 Tracer test. When a rise and fall tracer system is furnished, the following test shall be performed. Using an aluminum alloy workpiece not less than 3 inches wide and of suitable length and depth, and a slab mill of suitable design, the tracing capability shall be evaluated by trace-cutting the full length of the workpiece as follows: The workpiece shall be milled horizontally for its full length except that two plunge cuts shall be made, leaving each plunge cut between two horizontal milled surfaces. One of the plunge cuts

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shall be a tracing of two 20-degree slopes (fall and rise) 2.5 inches long with a two-inch horizontal cut between the slopes. The second plunge cut shall be an arc cut on a six-inch radius with a chord length of six inches between the horizontal segments. The reproduction of the workpiece shall be accurate to ±0.003 inch of all dimensions of the master template.

4.5.6 Electromagnetic interference control tests. Unless otherwise specified (see 6.2), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.14 in accordance with Table II of MIL-STD-461.

4.6 **Inspection of preparation for delivery**. Preservation, packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 preservation and Packaging. Preservation and packaging shall be in accordance with MIL-M-18058, at Level A or C, as specified (see 6.2).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B or C, as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2).

6. NOTES

6.1 Intended use The milling machines covered by this specification are to be used in the shops and depots of activities in the Department of Defense.

6.2 Ordering data. Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Special safety devices, if required (see 3.2.4).

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e. Painting, if different (see 3.3.5).

f. Power elevation and power rapid traverse for spindle carrier, when required (see 3.4.5).

g. Automatic quill retraction, when required (see 3.4.5.1).

h. Optional quill adjustment required, Option A or Option B (see 3.4.5.1).

i. Optional spindle speed required, Option A, B, C, or D (see 3.4.5.1).

j. Optional spindle horsepower required, Option A, B, C, D or E (see 3.4.5.1).

k. Overarms and supports for Type II machines, when required, Option A, B or C (see 3.4.6).

l. Vertical spindle travel and spindle distance of centerline of spindle to table top required, Option A, B or C (see 3.5. 1).

m. Electrical equipment, if different (see 3.8).

Type voltage characteristics of motor power supply, if different (see 3.8.1).

o. Voltage characteristics of initial wiring of machine (see 3.8.1).

P. Type of enclosure for motors, if different (see 3.8.1).

q. Electrical load indicator, if required (see 3.11).

r. Standard equipment requirement, if different (see 3.12).

s. Specify type optional equipment required (see 3.13).

t. Additional optional accessories and equipment (see 3.13.3).

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u. Electromagnetic interference control, if different (see 3.14 and 6,4).

v. Type fungus proofing, if required (see 3.15),

w. Lubrication chart or plate, if different (see 3.16),

x. Nameplate, if different (see 3,17).

y. Identification plate, if different (see 3.18).

z. First Article inspection, if different (see 4.2).

aa. Quality conformance inspection, if different (see 4.3).

bb. Electromagnetic interference control test requirements, if different (see 4.5.6 and 6.4).

cc. Required level of preservation and packaging (see 5,1).

dd. Required level of packing (see 5.2).

ee. Additional marking, if required (see 5.3).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.19),

6.3 First Article. When First Article approval is required under 3.1, the solicitation should include complete information concerning examination, test, and approval,

6.4 **Army and Air Force procurements.** Equipment purchased for the Departments of the Army and the Air Force should have electromagnetic interference control tests in conformance with MIL-STD-461, Table II, Class IIB equipment, except that only Test Methods CE03 and RE02 should be required.

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Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH, AS, OS

User Activities:
Army - SM, MU
Air Force
Navy - MC

Preparing Activity:
DSA-IP

Project Number:
3417-0080

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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